

Relation Between T-ratios

$$(i) \sin \theta = \frac{1}{\operatorname{cosec} \theta} \quad (ii) \cos \theta = \frac{1}{\sec \theta} \quad (iii) \tan \theta = \frac{1}{\cot \theta}$$

$$(iv) \tan \theta = \frac{\sin \theta}{\cos \theta} \quad (v) \cot \theta = \frac{\cos \theta}{\sin \theta}$$

EXAMPLE 5. If $\sin \theta = \frac{8}{17}$, then the other trigonometric ratios $\cos \theta$, $\tan \theta$, $\operatorname{cosec} \theta$, $\sec \theta$, $\cot \theta$ are

- a. $\frac{15}{17}, \frac{8}{15}, \frac{17}{8}, \frac{17}{15}, \frac{15}{8}$ b. $\frac{15}{17}, \frac{8}{15}, \frac{17}{8}, \frac{-17}{15}, \frac{8}{15}$
 c. $\frac{17}{15}, \frac{15}{8}, \frac{8}{17}, \frac{15}{17}, \frac{8}{15}$ d. None of these

Sol. a. As, $\sin \theta = \frac{8}{17} = \frac{\text{perpendicular}}{\text{hypotenuse}}$

In right angled ΔPQR ,

$$\therefore PQ^2 + QR^2 = PR^2 \Rightarrow PQ^2 = PR^2 - QR^2$$

$$\Rightarrow PQ = \sqrt{PR^2 - QR^2} = \sqrt{17^2 - 8^2}$$

$$= \sqrt{289 - 64} = \sqrt{225} = 15$$

$$\therefore \cos \theta = \frac{PQ}{PR} = \frac{15}{17}, \quad \tan \theta = \frac{RQ}{PQ} = \frac{8}{15}$$

$$\operatorname{cosec} \theta = \frac{PR}{RQ} = \frac{17}{8}, \quad \sec \theta = \frac{PR}{PQ} = \frac{17}{15}$$

$$\text{and } \cot \theta = \frac{PQ}{RQ} = \frac{15}{8}$$

EXAMPLE 6. If $\cot A = \frac{1}{(\sqrt{2}-1)}$, then the value of $\sin A \cos A$ is

- a. $\frac{1}{4}$ b. $\frac{1}{2}$ c. $\frac{1}{\sqrt{2}}$ d. $\frac{\sqrt{2}}{4}$

Sol. d. $\therefore \cot A = \frac{1}{\sqrt{2}-1} = \frac{\text{base}}{\text{perpendicular}}$

$$\Rightarrow \tan A = \frac{\sqrt{2}-1}{1} \left(\because \tan A = \frac{1}{\cot A} \right)$$

In right angled ΔABC ,

$$\therefore AC^2 = AB^2 + BC^2$$

$$\Rightarrow AC = \sqrt{AB^2 + BC^2}$$

$$= \sqrt{1^2 + (\sqrt{2}-1)^2} = \sqrt{1+2+1-2\sqrt{2}} = \sqrt{4-2\sqrt{2}}$$

$$\therefore \sin A = \frac{BC}{AC} = \frac{\sqrt{2}-1}{\sqrt{4-2\sqrt{2}}} \quad \text{and} \quad \cos A = \frac{AB}{AC} = \frac{1}{\sqrt{4-2\sqrt{2}}}$$

$$\therefore \sin A \cos A = \frac{\sqrt{2}-1}{\sqrt{4-2\sqrt{2}}} \cdot \frac{1}{\sqrt{4-2\sqrt{2}}} = \frac{\sqrt{2}-1}{4-2\sqrt{2}}$$

$$= \frac{\sqrt{2}-1}{2\sqrt{2}(\sqrt{2}-1)} = \frac{1}{2\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{\sqrt{2}}{4}$$

TRIGONOMETRIC IDENTITIES

A statement of equality involving trigonometric ratios of an angle is called a trigonometric identity. It is valid for all values of the angles. The three important identities are

- $\sin^2 \theta + \cos^2 \theta = 1$
- $1 + \tan^2 \theta = \sec^2 \theta$
- $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$

Note $\sin^2 A + \cos^2 B$ cannot be equal to 1 because the angles are different.

EXAMPLE 7. What is the value of $\frac{\tan A - \sin A}{\sin^3 A}$? **2013 I**

- a. $\frac{\sec A}{1 - \cos A}$ b. $\frac{\sec A}{1 + \cos^2 A}$
 c. $\frac{\sec A}{1 + \cos A}$ d. None of these

Sol. c. $\frac{\tan A - \sin A}{\sin^3 A} = \frac{\frac{\sin A}{\cos A} - \sin A}{\sin^3 A} = \frac{(1 - \cos A)}{\cos A \sin^2 A} \times \frac{(1 + \cos A)}{(1 + \cos A)}$

$$= \frac{1 - \cos^2 A}{\cos A \sin^2 A (1 + \cos A)} = \frac{\sin^2 A}{\cos A \sin^2 A (1 + \cos A)}$$

$$= \frac{1}{\cos A} \cdot \frac{1}{1 + \cos A} = \frac{\sec A}{1 + \cos A}$$

Values of Trigonometric Ratios for Some Specific Angles

Angles	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞
$\cot \theta$	∞	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
$\sec \theta$	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	∞
$\operatorname{cosec} \theta$	∞	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

EXAMPLE 8. What is the value of $\sin^3 60^\circ \cot 30^\circ - 2 \sec^2 45^\circ + 3 \cos 60^\circ \tan^2 45^\circ - \tan^2 60^\circ$?

- a. $\frac{35}{8}$ b. $\frac{-35}{8}$ c. $\frac{-11}{8}$ d. $\frac{11}{8}$

Sol. b. $\sin^3 60^\circ \cot 30^\circ - 2 \sec^2 45^\circ + 3 \cos 60^\circ \tan^2 45^\circ - \tan^2 60^\circ$

$$= \left(\frac{\sqrt{3}}{2} \right)^3 \times \sqrt{3} - 2 \times (\sqrt{2})^2 + 3 \times \frac{1}{2} \times (1)^2 - (\sqrt{3})^2$$

$$= \frac{9}{8} - 4 + \frac{3}{2} - 3 = \frac{9 - 32 + 12 - 24}{8} = \frac{21 - 56}{8} = \frac{-35}{8}$$

Sign of trigonometric ratio in different quadrants

II Quadrant $(90^\circ < \theta < 180^\circ)$ $\sin \theta$ and cosec θ are positive. All other trigonometric ratios are negative.	I Quadrant $(0 < \theta < 90^\circ)$ All trigonometric ratios are positive.
III Quadrant $(180^\circ < \theta < 270^\circ)$ $\tan \theta$ and cot θ are positive. All other trigonometric ratios are negative.	IV Quadrant $(270^\circ < \theta < 360^\circ)$ $\cos \theta$ and sec θ are positive. All other trigonometric ratios are negative.

EXAMPLE 9. If $0 < \theta < \frac{\pi}{4}$, then what is $\sqrt{1 - 2\sin\theta\cos\theta}$

equal to?

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- a. $\cos\theta - \sin\theta$ b. $\sin\theta - \cos\theta$
 c. $\pm(\cos\theta - \sin\theta)$ d. $\cos\theta\sin\theta$

Sol. a. Given, $0 < \theta < \frac{\pi}{4}$, then $\sqrt{1 - 2\sin\theta\cos\theta}$

$$\begin{aligned}
 &= \sqrt{\sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta} \quad [\because \sin^2\theta + \cos^2\theta = 1] \\
 &= \sqrt{(\cos\theta - \sin\theta)^2} \\
 &= \cos\theta - \sin\theta \quad \left[\because 0 < \theta < \frac{\pi}{4}, \cos\theta > \sin\theta, \text{ so we take } (\cos\theta - \sin\theta)^2 \right]
 \end{aligned}$$

Domain and Range of Trigonometrical Functions

Trigonometric Ratio	Domain	Range
$\sin \theta$	R	$[-1, 1]$
$\cos \theta$	R	$[-1, 1]$
$\tan \theta$	$R \sim \left\{ (2n+1)\frac{\pi}{2}, n \in I \right\}$	$(-\infty, \infty) = R$
$\cot \theta$	$R \sim \{n\pi, n \in I\}$	$(-\infty, \infty) = R$
$\sec \theta$	$R \sim \left\{ (2n+1)\frac{\pi}{2}, n \in I \right\}$	$(-\infty, -1] \cup [1, \infty)$
cosec θ	$R \sim \{n\pi, n \in I\}$	$(-\infty, -1] \cup [1, \infty)$

The behaviour of the trigonometric function in different quadrants are defined in following table.

Trigonometric Ratio	Ist Quadrant (θ increase from 0 to $\pi/2$)	IInd Quadrant (θ increase from $\pi/2$ to π)	III Quadrant (θ increase from π to $3\pi/2$)	IVth Quadrant (θ increase from $\frac{3\pi}{2}$ to 2π)
$\sin \theta$	Increases from 0 to 1	Decreases from 1 to 0	Decreases from 0 to -1	Increases from -1 to 0
$\cos \theta$	Decreases from 1 to 0	Decreases from 0 to -1	Increases from -1 to 0	Increases from 0 to 1
$\tan \theta$	Increases from 0 to ∞	Increases from $-\infty$ to 0	Increases from 0 to ∞	Increases from $-\infty$ to 0
$\cot \theta$	Decreases from ∞ to 0	Decreases from 0 to $-\infty$	Decreases from ∞ to 0	Decreases from 0 to $-\infty$
$\sec \theta$	Increases from 1 to ∞	Increases from $-\infty$ to -1	Decreases from -1 to $-\infty$	Decreases from ∞ to 1
cosec θ	Decreases from ∞ to 1	Increases from 1 to ∞	Increases from $-\infty$ to -1	Decreases from -1 to $-\infty$

EXAMPLE 10. Which one of the following statements is true in respect of the expression $\sin 31^\circ + \sin 32^\circ$?

- a. Its value is 0 b. Its value is 1
 c. Its value is less than 1 d. Its value is greater than 1

Sol. d. We know that, $\sin 30^\circ = \frac{1}{2}$

Value of $\sin$ increases 0° to 90°

$$\therefore \sin 31^\circ > \sin 30^\circ \text{ and } \sin 32^\circ > \sin 30^\circ$$

$$\Rightarrow \sin 31^\circ > \frac{1}{2} \text{ and } \sin 30^\circ > \frac{1}{2}$$

Adding both

$$\therefore \sin 31^\circ + \sin 31^\circ \Rightarrow \frac{1}{2} + \frac{1}{2}$$

$$\Rightarrow \sin 31^\circ + \sin 32^\circ > 1$$

IMPORTANT POINTS

- $|\sin \theta| \leq 1$
- $|\cos \theta| \leq 1$
- $\sec \theta \geq 1$
- $\sec \theta \leq -1$
- $\csc \theta \geq 1$
- $\csc \theta \leq -1$
- $\tan \theta$ and $\cot \theta$ can take any value.

TRIGONOMETRIC RATIOS OF ALLIED ANGLES

Two angles are said to be allied when their sum or difference is either zero or a multiple of 90° .

The angles like $-\theta, 90^\circ \pm \theta, 180^\circ \pm \theta, 360^\circ \pm \theta$ etc., are angles allied to the angle θ , if θ is measured in degrees.

Angle	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\cot \theta$	$\sec \theta$	cosec θ
$-\theta$	$-\sin \theta$	$\cos \theta$	$-\tan \theta$	$-\cot \theta$	$\sec \theta$	$-\csc \theta$
$90^\circ - \theta$	$\cos \theta$	$\sin \theta$	$\cot \theta$	$\tan \theta$	cosec θ	$\sec \theta$
$90^\circ + \theta$	$\cos \theta$	$-\sin \theta$	$-\cot \theta$	$-\tan \theta$	$-\csc \theta$	$\sec \theta$
$180^\circ - \theta$	$\sin \theta$	$-\cos \theta$	$-\tan \theta$	$-\cot \theta$	$-\sec \theta$	cosec θ
$180^\circ + \theta$	$-\sin \theta$	$-\cos \theta$	$\tan \theta$	$\cot \theta$	$-\sec \theta$	$-\csc \theta$
$270^\circ - \theta$	$-\cos \theta$	$\sin \theta$	$\cot \theta$	$\tan \theta$	$-\csc \theta$	$-\sec \theta$
$270^\circ + \theta$	$-\cos \theta$	$\sin \theta$	$-\cot \theta$	$-\tan \theta$	cosec θ	$-\sec \theta$
$360^\circ - \theta$	$-\sin \theta$	$\cos \theta$	$-\tan \theta$	$-\cot \theta$	$\sec \theta$	$-\csc \theta$
$360^\circ + \theta$	$\sin \theta$	$\cos \theta$	$\tan \theta$	$\cot \theta$	$\sec \theta$	cosec θ

EXAMPLE 11. What is the value of $\sec(90^\circ - \theta) \sin \theta \sec 45^\circ$?

- a. 1 b. $\frac{\sqrt{3}}{2}$ c. $\sqrt{2}$ d. $\sqrt{3}$

Sol. c. Given, $\sec(90^\circ - \theta) \sin \theta \sec 45^\circ = \operatorname{cosec} \theta \times \sin \theta \times \sqrt{2}$
 $= \frac{1}{\sin \theta} \times \sin \theta \times \sqrt{2} = \sqrt{2}$

EXAMPLE 12. The value of $\frac{\sin 135^\circ - \cos 120^\circ}{\sin 135^\circ + \cos 120^\circ}$ is

- a. $2 + 3\sqrt{2}$ b. $2 - 3\sqrt{2}$ c. $3 - 2\sqrt{2}$ d. $3 + 2\sqrt{2}$

Sol. d. $\sin 135^\circ = \sin(180^\circ - 45^\circ) = \sin 45^\circ = \frac{1}{\sqrt{2}}$
 $\cos 120^\circ = \cos(180^\circ - 60^\circ) = -\cos 60^\circ = -\frac{1}{2}$
 $\therefore \frac{\sin 135^\circ - \cos 120^\circ}{\sin 135^\circ + \cos 120^\circ} = \frac{\frac{1}{\sqrt{2}} - \left(-\frac{1}{2}\right)}{\frac{1}{\sqrt{2}} + \left(-\frac{1}{2}\right)} = \frac{\frac{\sqrt{2} + 1}{2}}{\frac{\sqrt{2} - 1}{2}} \times \frac{\sqrt{2} + 1}{\sqrt{2} + 1}$
 $= \frac{(\sqrt{2} + 1)^2}{2 - 1} = 2 + 1 + 2\sqrt{2} = 3 + 2\sqrt{2}$

Sum, Difference and Product Formulae

- $\sin(A + B) = \sin A \cos B + \cos A \sin B$
- $\sin(A - B) = \sin A \cos B - \cos A \sin B$
- $\cos(A + B) = \cos A \cos B - \sin A \sin B$
- $\cos(A - B) = \cos A \cos B + \sin A \sin B$
- $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$
- $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
- $\cot(A + B) = \frac{\cot A \cot B - 1}{\cot A + \cot B}$
- $\cot(A - B) = \frac{\cot A \cot B + 1}{\cot B - \cot A}$
- $\sin(A + B) \sin(A - B) = \sin^2 A - \sin^2 B$
 $= \cos^2 B - \cos^2 A$
- $\cos(A + B) \cos(A - B) = \cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A$
- $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$
- $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$
- $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$
- $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$
- $\sin A + \sin B = 2 \sin \frac{(A + B)}{2} \cdot \cos \frac{(A - B)}{2}$

- $\sin A - \sin B = 2 \sin \frac{(A - B)}{2} \cdot \cos \frac{(A + B)}{2}$
- $\cos A + \cos B = 2 \cos \frac{(A + B)}{2} \cdot \cos \frac{(A - B)}{2}$
- $\cos A - \cos B = -2 \sin \frac{(A + B)}{2} \cdot \sin \frac{(A - B)}{2}$
 $= 2 \sin \frac{(A + B)}{2} \cdot \sin \frac{(B - A)}{2}$
- $\tan A + \tan B = \frac{\sin(A + B)}{\cos A \cos B}$
- $\tan A - \tan B = \frac{\sin(A - B)}{\cos A \cos B}$
- $\cot A + \cot B = \frac{\sin(A + B)}{\sin A \sin B}$
- $\cot A - \cot B = \frac{\sin(B - A)}{\sin A \sin B}$

EXAMPLE 13. If $A - B = \frac{\pi}{3}$, then the value of $\cos A \cos B + \sin A \sin B$ is

- a. $1/2$ b. 1
c. $3/2$ d. None of these

Sol. a. $\cos A \cos B + \sin A \sin B = \cos(A - B)$

$$= \cos\left(\frac{\pi}{3}\right) = \frac{1}{2} \quad \left(\because A - B = \frac{\pi}{3}\right)$$

EXAMPLE 14. The value of $\frac{\sqrt{3} \cos 23^\circ - \sin 23^\circ}{2}$ is

- a. $\cos 7^\circ$ b. $\sin 53^\circ$ c. $\cos 53^\circ$ d. $\sin 7^\circ$

Sol. c. $\frac{\sqrt{3} \cos 23^\circ - \sin 23^\circ}{2} = \frac{\sqrt{3}}{2} \cos 23^\circ - \frac{1}{2} \sin 23^\circ$
 $= \cos 30^\circ \cos 23^\circ - \sin 30^\circ \sin 23^\circ$
 $= \cos(30^\circ + 23^\circ) = \cos 53^\circ$
 $[\because \cos(A + B) = \cos A \cos B - \sin A \sin B]$

EXAMPLE 15. The value of $\frac{\tan 47^\circ + \tan 43^\circ}{1 - \tan 47^\circ \tan 43^\circ}$ is

- a. 1 b. ∞
c. 0 d. -1

Sol. b. $\therefore \frac{\tan A + \tan B}{1 - \tan A \tan B} = \tan(A + B) \Rightarrow \frac{\tan 47^\circ + \tan 43^\circ}{1 - \tan 47^\circ \tan 43^\circ}$
 $= \tan(47^\circ + 43^\circ) = \tan 90^\circ = \infty$

EXAMPLE 16. The value of $\sin 20^\circ \sin 40^\circ \sin 60^\circ \sin 80^\circ$ is

- a. $\frac{1}{2}$ b. $\frac{3}{4}$ c. $\frac{1}{8}$ d. $\frac{3}{16}$

Sol. d. $\sin 20^\circ \sin 40^\circ \sin 80^\circ \sin 60^\circ$

$$\begin{aligned}
 &= \sin 60^\circ [\sin 80^\circ \sin 40^\circ] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{2} [\sin 80^\circ \sin 40^\circ] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{2} \times \frac{1}{2} [2 \sin 80^\circ \sin 40^\circ] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{4} [\cos(80^\circ - 40^\circ) - \cos(80^\circ + 40^\circ)] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{4} [\cos 40^\circ - \cos 120^\circ] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{4} \left[\cos 40^\circ - \left(-\frac{1}{2}\right) \right] \sin 20^\circ \quad \left(\because \cos 120^\circ = -\frac{1}{2} \right) \\
 &= \frac{\sqrt{3}}{4} \left[\cos 40^\circ + \frac{1}{2} \right] \sin 20^\circ \\
 &= \frac{\sqrt{3}}{8} [\sin(40^\circ + 20^\circ) - \sin(40^\circ - 20^\circ) + \sin 20^\circ] \\
 &= \frac{\sqrt{3}}{8} [\sin 60^\circ - \sin 20^\circ + \sin 20^\circ] \\
 &= \frac{\sqrt{3}}{8} \times \sin 60^\circ = \frac{\sqrt{3}}{8} \times \frac{\sqrt{3}}{2} = \frac{3}{16}
 \end{aligned}$$

EXAMPLE 17. $\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x}$ is equal to

- a. $\cot 3x$ b. $\tan 3x$ c. $\cot x$ d. $\cot 2x$

Sol. a. $\frac{\cos 4x + \cos 3x + \cos 2x}{\sin 4x + \sin 3x + \sin 2x} = \frac{(\cos 4x + \cos 2x) + \cos 3x}{(\sin 4x + \sin 2x) + \sin 3x}$

$$\begin{aligned}
 &= \frac{2 \cos \left(\frac{4x+2x}{2} \right) \cos \left(\frac{4x-2x}{2} \right) + \cos 3x}{2 \sin \left(\frac{4x+2x}{2} \right) \cos \left(\frac{4x-2x}{2} \right) + \sin 3x} \\
 &= \frac{2 \cos 3x \cdot \cos x + \cos 3x}{2 \sin 3x \cdot \cos x + \sin 3x} = \frac{\cos 3x (2 \cos x + 1)}{\sin 3x (2 \cos x + 1)} = \cot 3x
 \end{aligned}$$

Trigonometric Ratios of Multiple Angles

- $\sin 2\theta = 2 \sin \theta \cos \theta = \frac{2 \tan \theta}{1 + \tan^2 \theta}$
- $\cos 2\theta = \cos^2 \theta - \sin^2 \theta = 2 \cos^2 \theta - 1$
 $= 1 - 2 \sin^2 \theta = \frac{1 - \tan^2 \theta}{1 + \tan^2 \theta}$
- $\frac{1 + \cos 2\theta}{2} = \cos^2 \theta$ and $\frac{1 - \cos 2\theta}{2} = \sin^2 \theta$
- $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ • $\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$
- $\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$ • $\tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$

EXAMPLE 18. The value of $3 \sin 15^\circ - 4 \sin^3 15^\circ$ is

- a. 1 b. $\frac{1}{2}$ c. $\frac{1}{\sqrt{2}}$ d. 0

Sol. c. As, $3 \sin \theta - 4 \sin^3 \theta = \sin 3\theta$

$$\text{So, } 3 \sin 15^\circ - 4 \sin^3 15^\circ = \sin 3(15^\circ) = \sin 45^\circ = \frac{1}{\sqrt{2}}$$

Trigonometric Ratios of Submultiple Angles

- $\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}$
- $\cos \theta = \cos^2 \frac{\theta}{2} - \sin^2 \frac{\theta}{2} = 2 \cos^2 \frac{\theta}{2} - 1 = 1 - 2 \sin^2 \frac{\theta}{2}$
- $\frac{1 - \cos \theta}{\sin \theta} = \tan \frac{\theta}{2}$ • $\frac{1 + \cos \theta}{\sin \theta} = \cot \frac{\theta}{2}$
- $\frac{1 - \cos \theta}{1 + \cos \theta} = \tan^2 \frac{\theta}{2}$ • $\frac{1 + \cos \theta}{1 - \cos \theta} = \cot^2 \frac{\theta}{2}$
- $\sin \theta = \frac{2 \tan \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$ • $\cos \theta = \frac{1 - \tan^2 \frac{\theta}{2}}{1 + \tan^2 \frac{\theta}{2}}$

Maximum and Minimum Values of Trigonometrical Functions

We know that, $-1 \leq \sin \theta \leq 1$, $-1 \leq \cos \theta \leq 1$

Maximum and minimum value of a trigonometrical function of the form $a \sin \theta \pm b \cos \theta$ are $\sqrt{a^2 + b^2}$

and $-\sqrt{a^2 + b^2}$, respectively.

EXAMPLE 19. The maximum value and minimum value of $8 \sin \theta \cos \theta + 4 \cos 2\theta$ is

- a. 4 and -4 b. 16 and -16
 c. $4\sqrt{2}$ and $-4\sqrt{2}$ d. None of these

Sol. c. Here, $8 \sin \theta \cos \theta + 4 \cos 2\theta = 4 \sin 2\theta + 4 \cos 2\theta$

$$\text{So, maximum value} = \sqrt{4^2 + 4^2} = 4\sqrt{2}$$

$$\text{and minimum value} = -\sqrt{4^2 + 4^2} = -4\sqrt{2}$$

EXAMPLE 20. Solve $\cos \theta + \sin \theta = \sqrt{2}$, then the value of θ is

- a. $\pi/3$ b. $\pi/4$ c. $\pi/6$ d. $\pi/2$

Sol. b. $\cos \theta + \sin \theta = \sqrt{2}$, $\frac{\cos \theta}{\sqrt{2}} + \frac{\sin \theta}{\sqrt{2}} = 1$
 ($\because$ dividing throughout by $\sqrt{2}$)
 $\Rightarrow \cos \theta \cos 45^\circ + \sin \theta \sin 45^\circ = 1$
 $\cos(\theta - 45^\circ) = 1 = \cos 0^\circ$
 $\therefore \theta - 45^\circ = 0 \Rightarrow \theta = 45^\circ = \pi/4$

EXAMPLE 21. The value of ϕ for maximum value of $\sin 3\phi + \cos 3\phi$ is

- a. $\sqrt{2}$ b. 90° c. 1 d. 15°

Sol. d. $\sin 3\phi + \cos 3\phi = \sqrt{2} \left(\frac{1}{\sqrt{2}} \cos 3\phi + \frac{1}{\sqrt{2}} \sin 3\phi \right)$
 $= \sqrt{2} (\sin 45^\circ \cos 3\phi + \cos 45^\circ \sin 3\phi)$
 $= \sqrt{2} \sin (45^\circ + 3\phi)$

$\therefore$ The maximum value occurs when

$$\sin (45^\circ + 3\phi) = 1 = \sin 90^\circ$$

$$\therefore 45^\circ + 3\phi = 90^\circ \Rightarrow \phi = 15^\circ$$

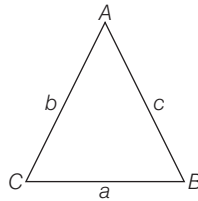
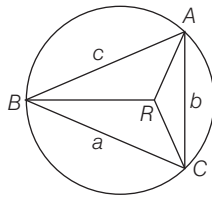
Properties of Triangle

1. **Circumcircle of a triangle** The circle passing through the vertices of a $\triangle ABC$ is called circumcircle. Its radius R is called the circumradius and its centre is known as circumcentre.

$$\text{Here, } R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

2. **Sine rule** In a $\triangle ABC$, if a, b, c be the three sides opposite to the angles A, B, C respectively, then

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



3. **Cosine rule** In a $\triangle ABC$, if a, b, c be the sides opposite to angles A, B and C respectively, then

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \cos B = \frac{c^2 + a^2 - b^2}{2ac} \text{ and}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

EXAMPLE 22. A triangle ABC is inscribed in a circle. If sum of the squares of sides of the triangle is equal to twice the square of the diameter, then $\sin^2 A + \sin^2 B + \sin^2 C$ is equal to

- a. 2 b. 3 c. 4 d. None of these

Sol. a. Let the radius of inscribed circle be R .

$$\therefore R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$$

$$\Rightarrow \sin A = \frac{a}{2R}, \sin B = \frac{b}{2R} \text{ and } \sin C = \frac{c}{2R}$$

$$\sin^2 A + \sin^2 B + \sin^2 C = \frac{a^2 + b^2 + c^2}{4R^2}$$

$$\Rightarrow \sin^2 A = \frac{a^2}{4R^2}, \sin^2 B = \frac{b^2}{4R^2} \text{ and } \sin^2 C = \frac{c^2}{4R^2}$$

According to the question,

$$a^2 + b^2 + c^2 = 2 \times 2R^2 = 8R^2$$

$$\therefore \sin^2 A + \sin^2 B + \sin^2 C = \frac{8R^2}{4R^2} = 2$$

> PRACTICE EXERCISE

- A unit radian is approximately equal to
 (a) $57^\circ 17' 43''$ (b) $57^\circ 16' 22''$
 (c) $57^\circ 17' 47''$ (d) $57^\circ 17' 49''$
- Find the radian measure corresponding to the degree $-47^\circ 30'$.
 (a) $-\frac{19\pi}{72}$ rad (b) $\frac{17\pi}{72}$ rad
 (c) $\frac{13\pi}{72}$ rad (d) $-\frac{15\pi}{72}$ rad
- The length of a pendulum is 60 cm. The angle through which it swings when its tip describes an arc of length 16.5 cm is
 (a) $15^\circ 30'$ (b) $15^\circ 45'$ (c) $16^\circ 15'$ (d) $16^\circ 45'$
- A railway train is moving on a circular curve of radius 1500 m at a speed of 90 km/h. Through what angle has it turned in 11 seconds?
 (a) 12° (b) $16^\circ 30'$ (c) $10^\circ 30'$ (d) $11^\circ 40'$
- The angle between the minute hand and the hour hand of a clock when the time is 7:35 pm is
 (a) $16^\circ 45'$ (b) $17^\circ 30'$ (c) $18^\circ 15'$ (d) $19^\circ 30'$
- When do the hands of a clock coincide between 5 pm and 6 pm?
 (a) 5:30 pm (b) 5:27:16 pm
 (c) 5:32:16 pm (d) 5:28:56 pm
- If $\sin A = \frac{1}{3}$, then $\cos A \cdot \operatorname{cosec} A + \tan A \cdot \sec A$ is equal to
 (a) $\frac{16\sqrt{2} + 3}{8}$ (b) $\frac{4\sqrt{2} + 3}{8}$
 (c) $\frac{\sqrt{3} + 2}{4}$ (d) $\frac{\sqrt{3} - 1}{8}$
- The value of $\cos 15^\circ - \sin 15^\circ$ is equal to
 (a) $\frac{1}{3}$ (b) $\frac{1}{\sqrt{2}}$ (c) $\frac{1}{2}$ (d) $\frac{\sqrt{3}}{2}$